DEGREE FILTRATION BY OPEN SUBSPACES FOR FLAT FINITELY PRESENTED SEPARATED LOCALLY QUASI-FINITE MORPHISMS

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Editorial note. The source marks base-change commutation 'Clearly'; carried as an explicit obligation.

Allowed standard prerequisites (editorial).

- Scheme-side degree filtration. More on Morphisms, lemma stratify-flat-fp-lqf (cross-chapter, named). Produces the filtration of U.
- Descending open subspaces through presentations. Properties of Spaces, lemma subspaces-presentation (cross-chapter, named). Descends the R-invariant opens U_d to X_d.

Source reference. Original reference 086R: focused-seeds/geometry/S9-decent-stratify-flat-fp-lqf/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07TW: focused-seeds/geometry/S9-decent-stratify-flat-fp-lqf/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $W \to X$ be a morphism of a scheme W to an algebraic space X which is flat, locally of finite presentation, separated, and locally quasi-finite. Then there exist open subspaces*

$$X = X_0 \supset X_1 \supset X_2 \supset \ldots$$

such that a morphism $\mathrm{Spec}(k) \to X$ where k is a field factors through X_d if and only if $W \times_X \mathrm{Spec}(k)$ has degree $\geq d$ over k.

Proof. Choose a scheme U and a surjective étale morphism $U \to X$. Apply More on Morphisms, Lemma 086R to $W \times_X U \to U$. We obtain open subschemes

$$U = U_0 \supset U_1 \supset U_2 \supset \ldots$$

characterized by the property stated in the lemma for the morphism $W \times_X U \to U$. Clearly, the formation of these closed subsets commutes with base change. Setting $R = U \times_X U$ with projection maps $s, t : R \to U$ we conclude that

$$s^{-1}(U_d) = t^{-1}(U_d)$$

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as open subschemes of R. In other words the open subschemes $U_d \subset U$ are R-invariant. This means that U_d is the inverse image of an open subspace $X_d \subset X$ (Properties of Spaces, Lemma 07TW). $\square$